

Technical Indicators & Strategy Backtesting

FINM 25000 – Assignment 2

AMD — 2020-01-01 to 2025-01-01 — Daily, split/dividend adjusted

1 Strategies

Strategy 1 — Trend Following (MACD)

Category: Trend. This strategy uses the MACD line (EMA12 – EMA26) and its 9-day signal line.

- **Entry:** buy when the MACD line goes above the signal line.
- **Exit:** sell when the MACD line goes below the signal line.

Strategy 2 — Mean Reversion (RSI + Bollinger Bands)

Categories: Momentum and Volatility. This strategy uses the 14-day RSI and the 20-day Bollinger Bands.

- **Entry:** buy when $RSI < 30$ and the close price is below the lower Bollinger Band (the stock is oversold).
- **Exit:** sell when $RSI > 70$ and the close price is above the upper Bollinger Band (the stock is overbought).

Strategy 3 — Custom (EMA + ATR + OBV)

Categories: Trend, Volatility, and Volume (three categories). This strategy uses three signals together: the EMA20 and EMA50 trend, the ATR and its 20-day average, and the OBV and its 20-day average.

- **Entry:** buy when $EMA20 > EMA50$ (up trend) AND $ATR > \text{its average}$ (the range is bigger) AND $OBV > \text{its average}$ (the volume agrees).
- **Exit:** sell when the trend breaks ($EMA20 < EMA50$) OR the volume gets weak ($OBV < \text{its average}$) OR the volatility gets small ($ATR < \text{its average}$).

2 Performance Comparison

Strategy	Total Ret.	CAGR	Vol.	Sharpe	Sortino	Max DD	Win Rate
Buy & Hold	146.0%	19.8%	52.4%	0.61	0.93	−65.4%	n/a
Trend Following	172.9%	22.3%	37.8%	0.72	0.83	−40.7%	40.9%
Mean Reversion	8.2%	1.6%	30.5%	0.20	0.17	−57.9%	66.7%
Custom	192.2%	24.0%	24.8%	0.99	0.95	−29.0%	42.9%

3 Discussion

Apple is a company with stable product demand. So I did not pick Apple. I picked AMD instead. AMD's product demand changes more, so its price also moves more. I think this kind of stock is better for trend following.

From the table, the Custom strategy is the best on a risk-adjusted basis. It has the highest Sharpe (0.99) and the highest Sortino (0.95). It also has the highest total return (192%) and the smallest max drawdown (−29%). This strategy only buys when the trend, the volatility, and the volume all agree. Because of this, it stays out of the market during the big AMD drop in 2022. This is the time when Buy & Hold loses the most (−65%).

Trend Following is also better than Buy & Hold. It returns 173% and Buy & Hold returns 146%. It also has a higher Sharpe and about half the drawdown. This shows that a simple MACD rule can help for a stock that moves a lot like AMD. Mean Reversion is the worst one. It only returns 8%. It does not trade often, and it misses most of the up trend. But it has the highest win rate (67%). So it wins many small trades, but it loses a big chance to make money.

Buy & Hold returns 146%, but its risk is the worst (Sharpe 0.61, −65% drawdown). It has no win rate because it never sells. Also, I did not add trading fees and slippage. If I add these costs, the strategies that trade many times will become a little worse.

I have two more points. First, this backtest is long-only. So when a strategy sells AMD, the money just stays in cash and earns nothing. But in real life the money is not stuck. I can short AMD to make money when the price goes down, or I can buy another stock. Both ways can make more profit during the flat time. So the real chance to make money is bigger than these numbers show.

Second, these strategies are made for one company. The indicators and the parameters fit AMD's own price behavior. The same rules may work very differently on another stock. So these results only show how the strategies work on AMD in this period. They do not mean one strategy is the best for every stock.